

The abdominal wall

Department of Anatomy

The abdomen (1)

First Part

Dr. Osman Ahmed Osman

Lecture points

- **Location of Abdomen.**
- **The abdomen region in anatomical base**
- **The superior aperture of the abdomen.**
- **The component parts of abdominal walls.**
- **Pelvic Inlet.**
- **Relationship of the regions to the abdomen.**
- **The abdominal cavity is continuous with the pelvic cavity.**
- **The component parts of abdominal walls.**
- **Abdominal Cavity**
- **The abdominal Viscera**
- **The peritoneal cavity and peritoneum**
- **Arterial supply of abdominal organs**
- **Abdomen assist in breathing**
- **Posterior Abdominal Wall**
- **Boundaries of the Abdomen**
- **Anatomical landmark**
- **The division of abdomen into four quadrants**
- **The nine region of the abdomen.**
- **The Anterior Abdominal Wall Functions.**
- **The Layers of anterior abdominal wall.**
- **Transvers section showing the layers of the abdominal wall.**
- **The Superficial fascia.**
- **Associated ligaments.**
- **Muscles of the anterior abdominal wall.**
- **The lumbar triangle.**
- **Rectus sheath.**
- **The inguinal ligament.**
- **Cutaneous innervation Of the anterior abdominal wall.**
- **Blood vessels of the anterior abdominal wall**
- **Venous Drainage.**
- **Lymphatic Drainage**

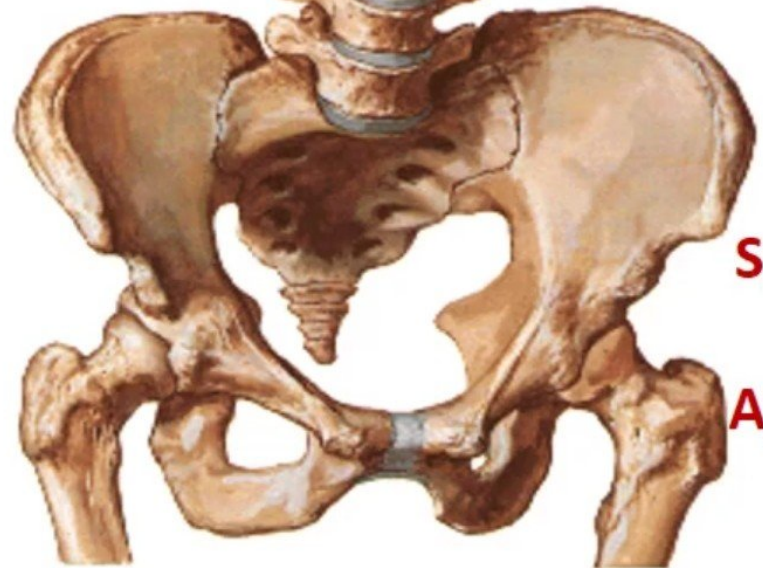
Location of Abdomen

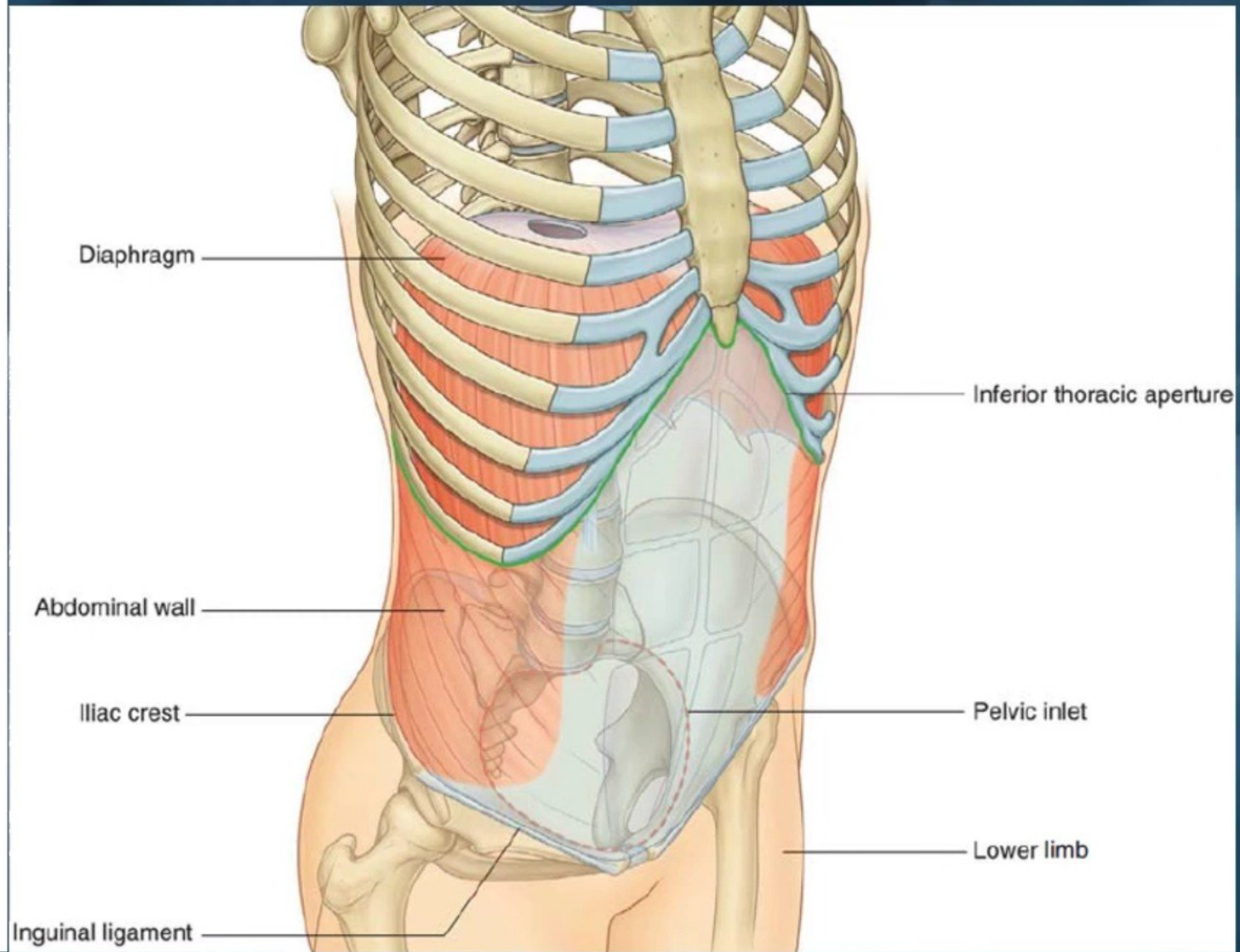
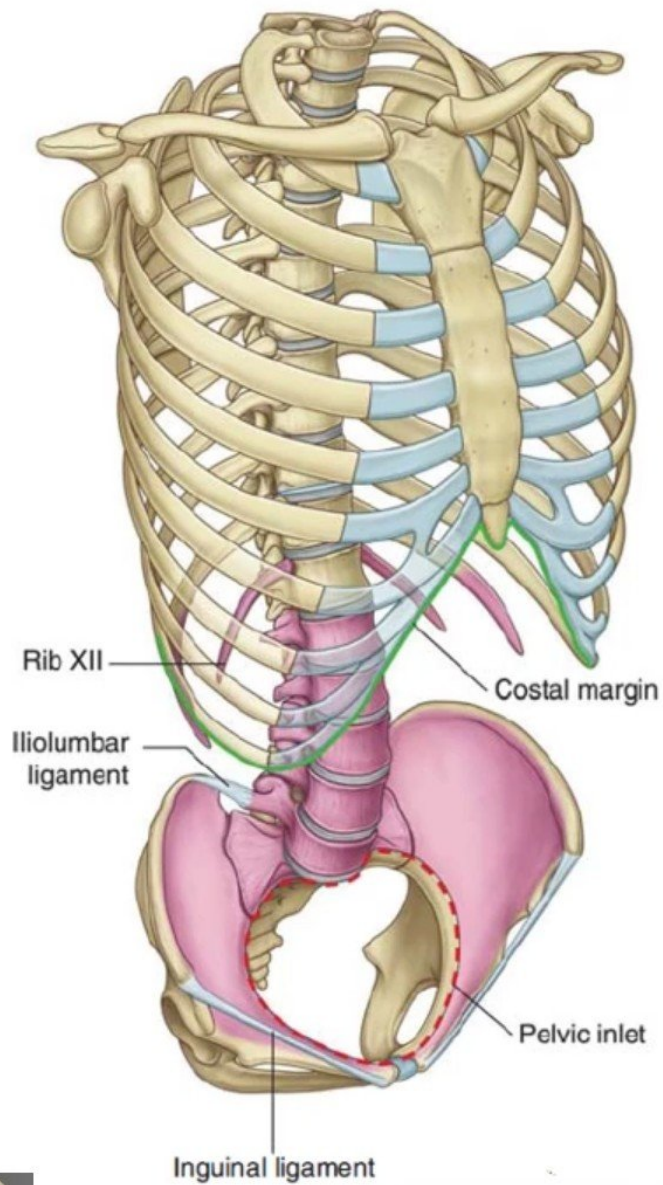
- The abdomen is Cylindrical chamber.
- The inferior thoracic aperture forms the superior opening to the abdomen.
 - ✓ And is closed by the diaphragm.
- The superior pelvic aperture forms the inferior opening of the abdomen
 - ✓ and the superior margin of the lower limb.

Inferior
Thoracic
Aperture



Superior
Pelvic
Aperture



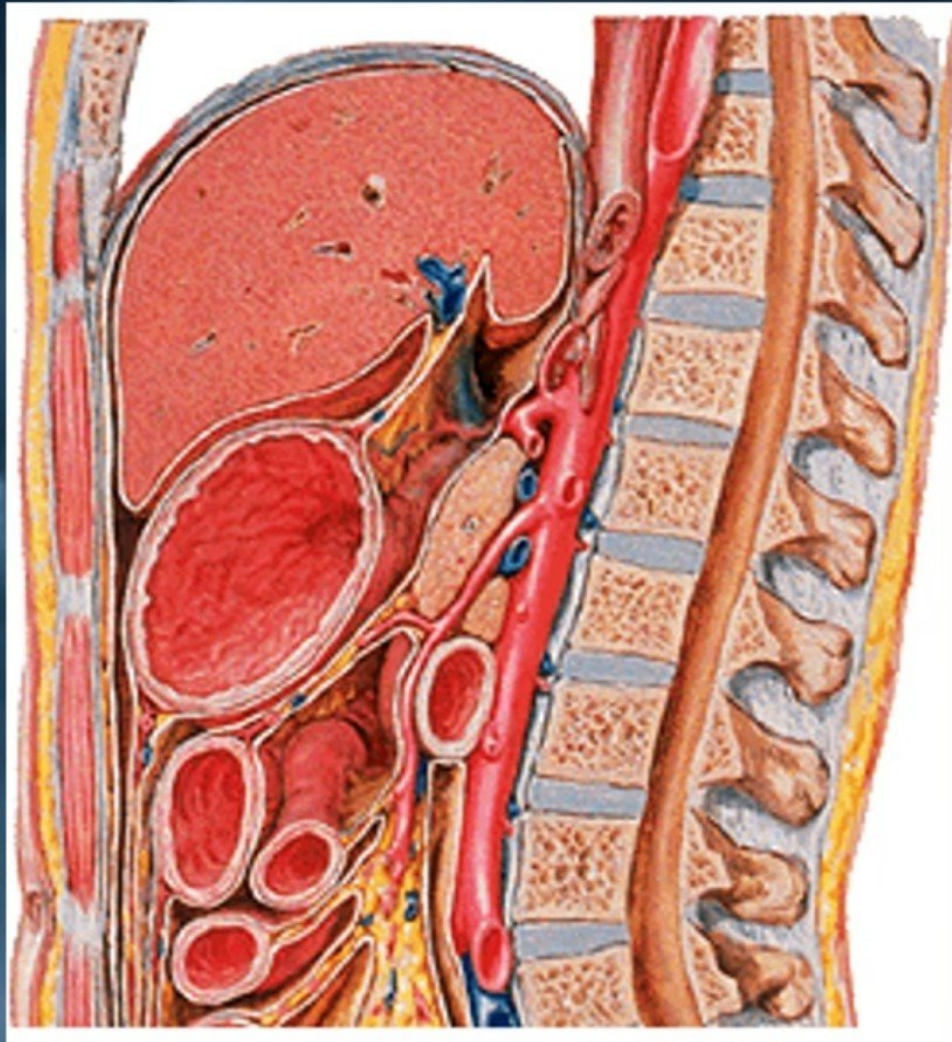


The abdomen region in anatomical base

Note:

The abdominal cavity and the abdominal wall enclosed by peritoneum to form the abdominal cavity.

Anterior
Abdominal
Wall

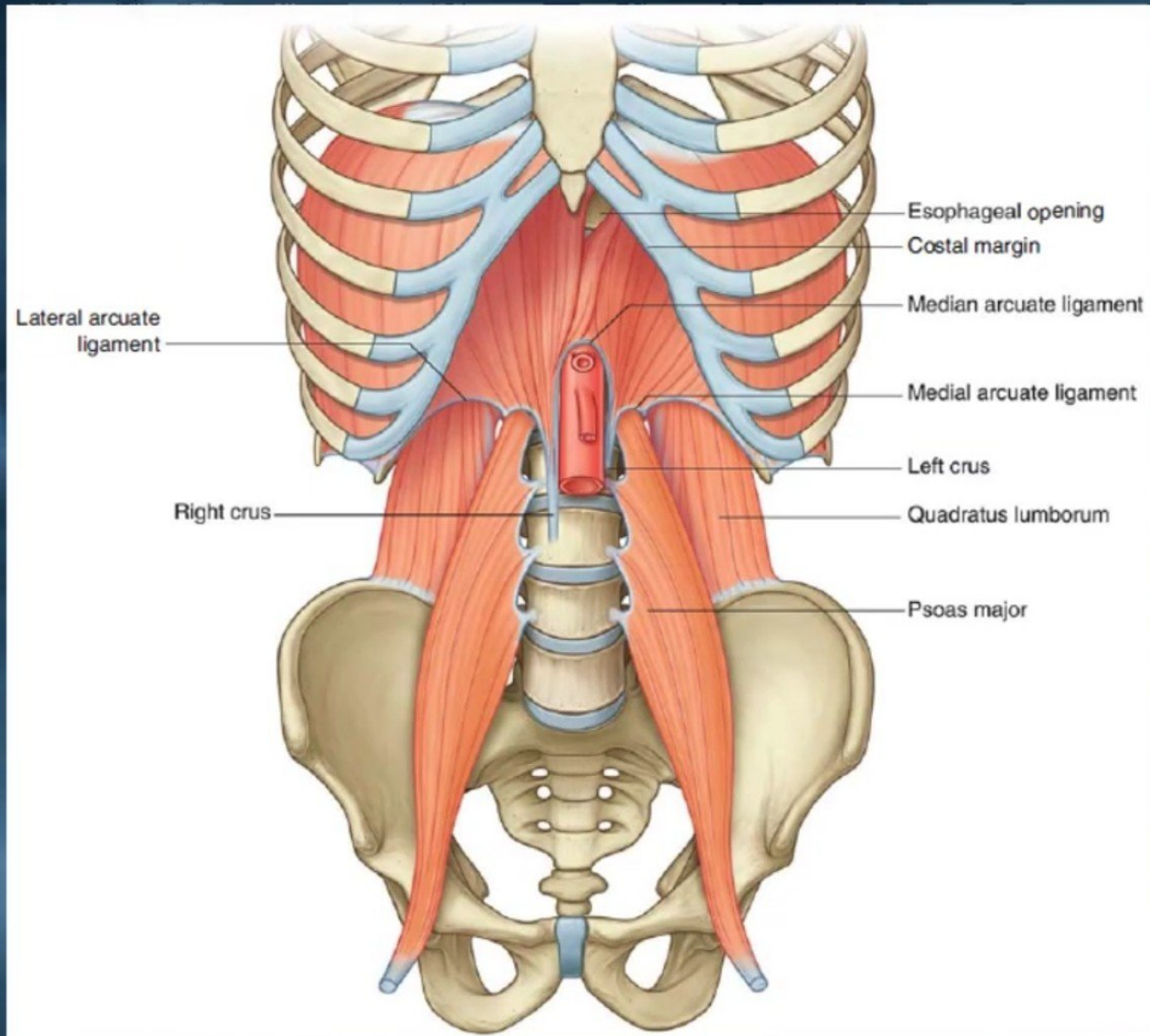


Posterior
Abdominal
Wall

Abdominal
Cavity

The superior aperture of the abdomen

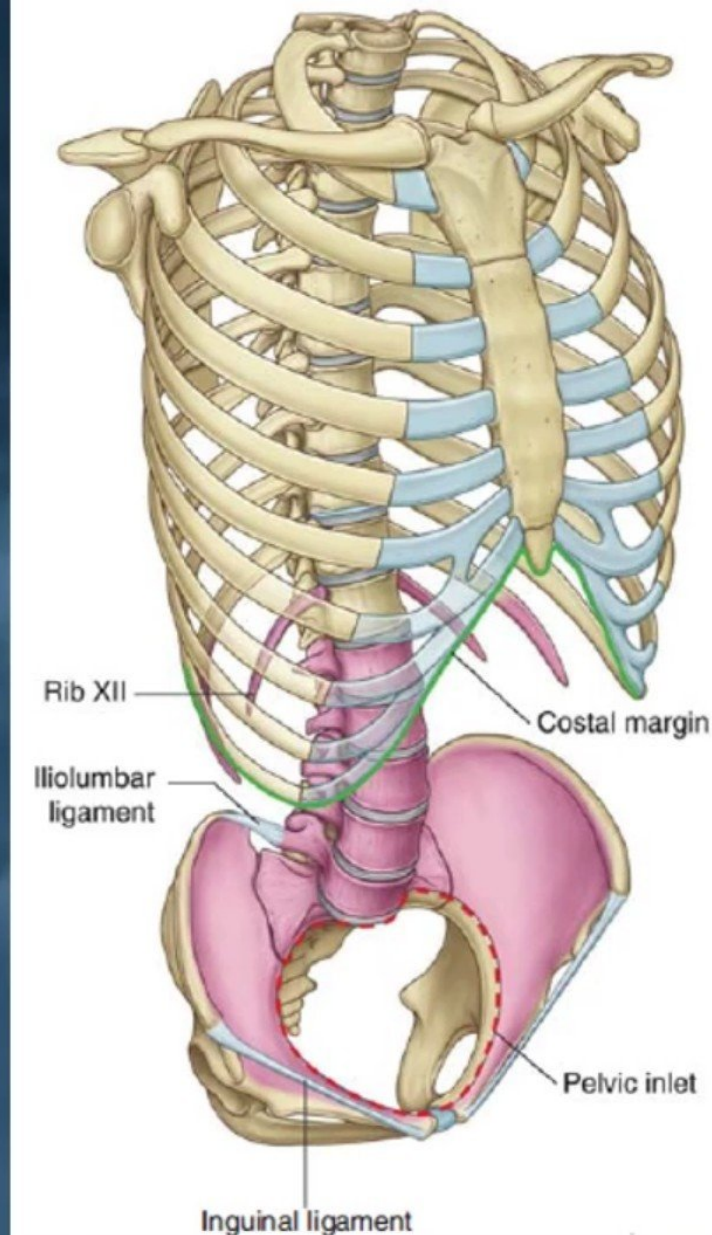
The superior aperture of the abdomen is **the inferior thoracic aperture**, which is closed by the **diaphragm**.



The component parts of abdominal walls

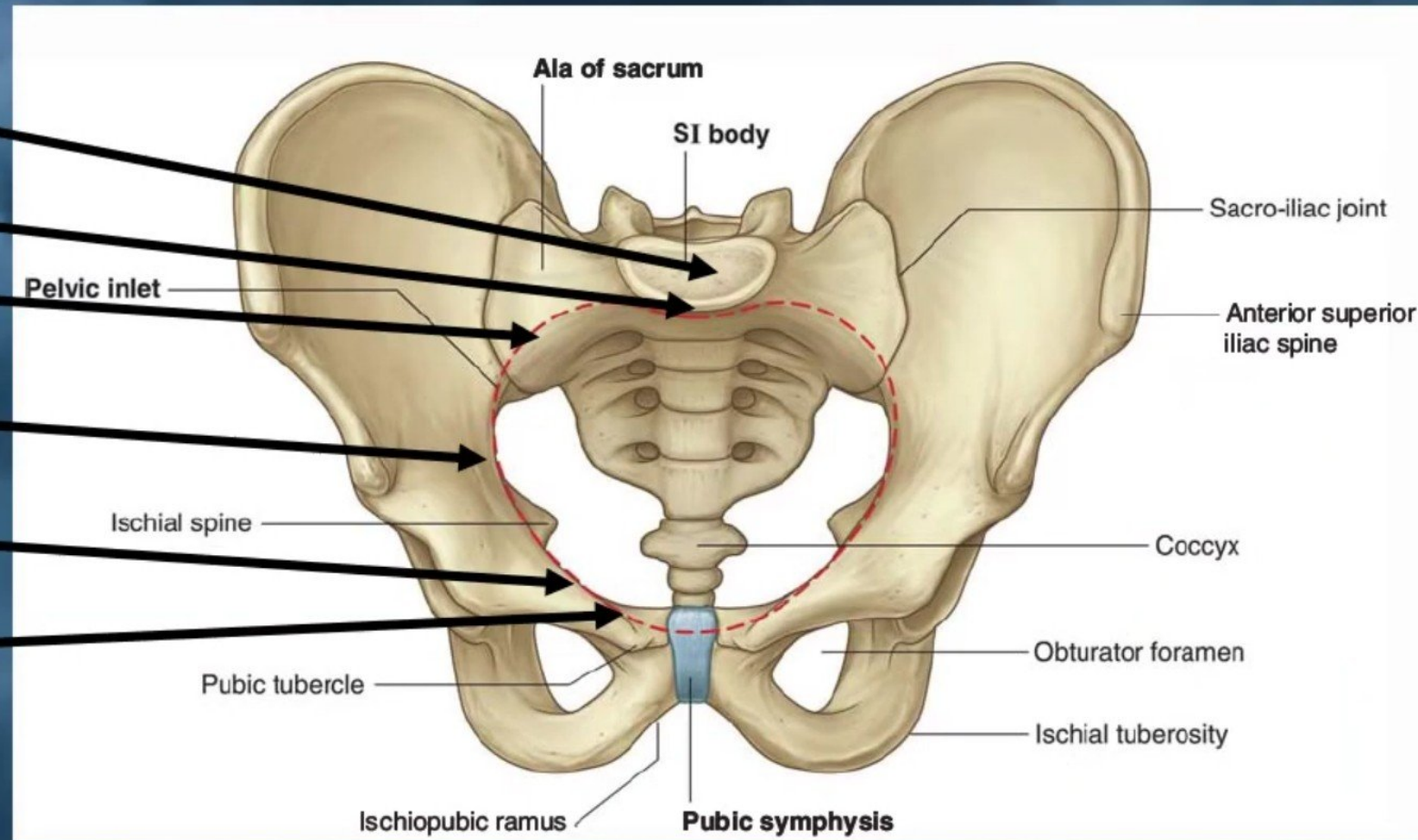
The skeletal elements of the walls:

- **Posteriorly found:** The **five lumbar vertebrae** and their interesting **intervertebral discs**.
- **Inferiorly found:** The superior expanded parts of **the pelvic bone** (from pelvic inlet).
- The bony components of **the inferior thoracic wall**, including: **the lower border of costal margin, ribs XII, XI and Xiphoid process**.



Pelvic Inlet

- The body of the vertebra S1.
- The sacral promontory.
- On each side of the vertebra found the ala (wings) of sacrum.
- Arcuate line of the ilium.
- Pectineal line of the pubis.
- The pubic crest.



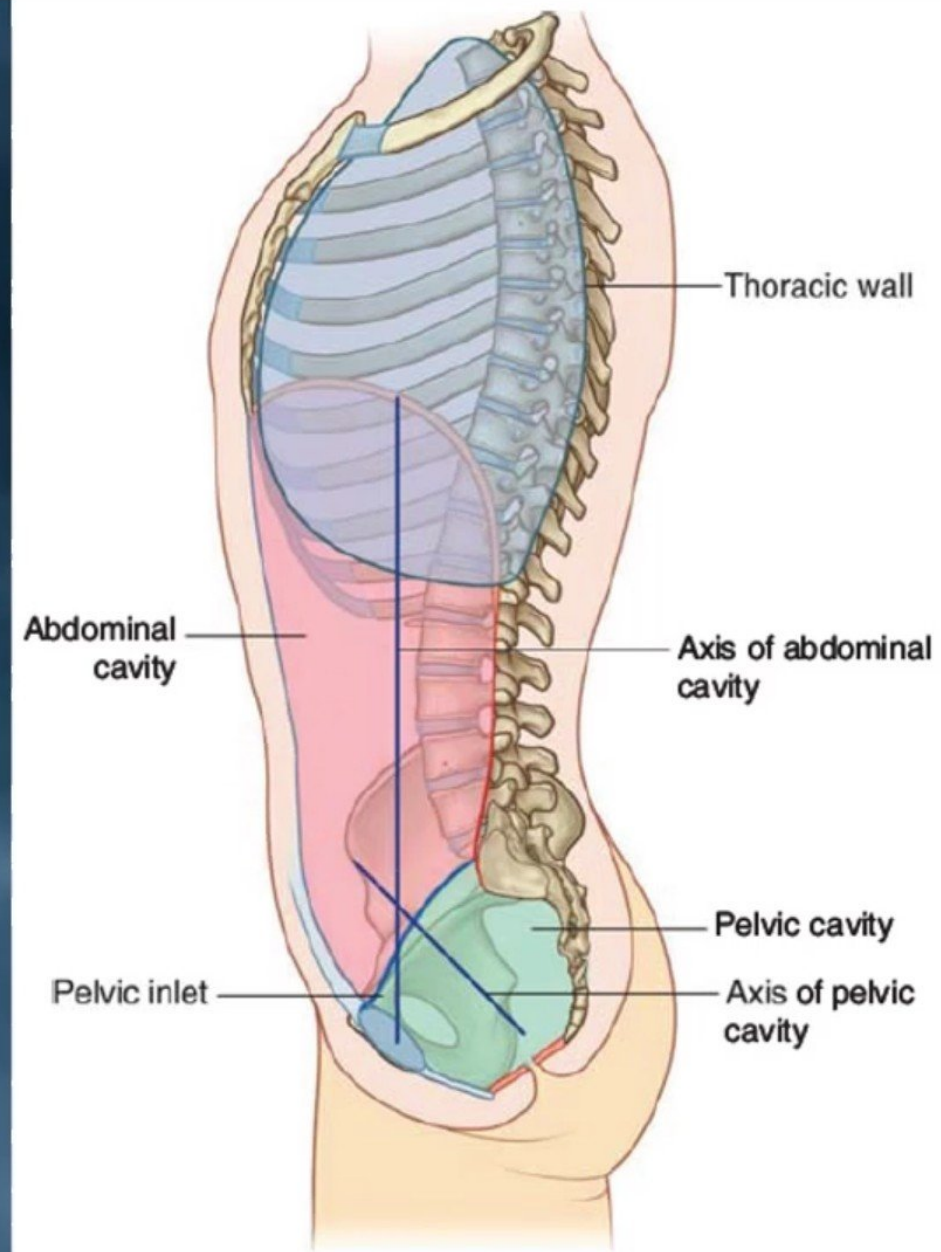
Relationship of the regions to the abdomen

Thorax:

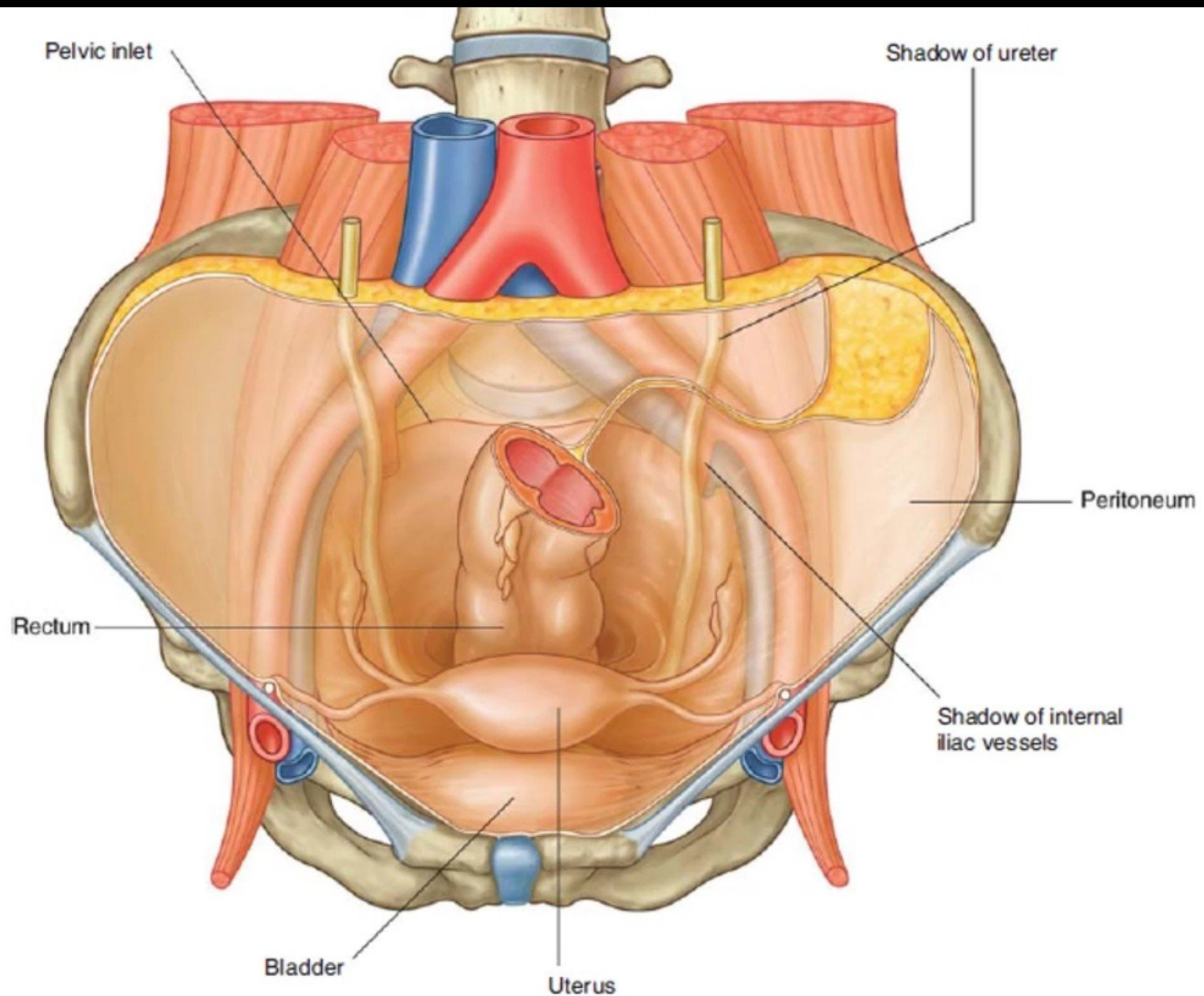
- The abdomen is separated from the thorax by the diaphragm.
- The structures pass between the two regions through or posterior to the diaphragm.

Pelvis:

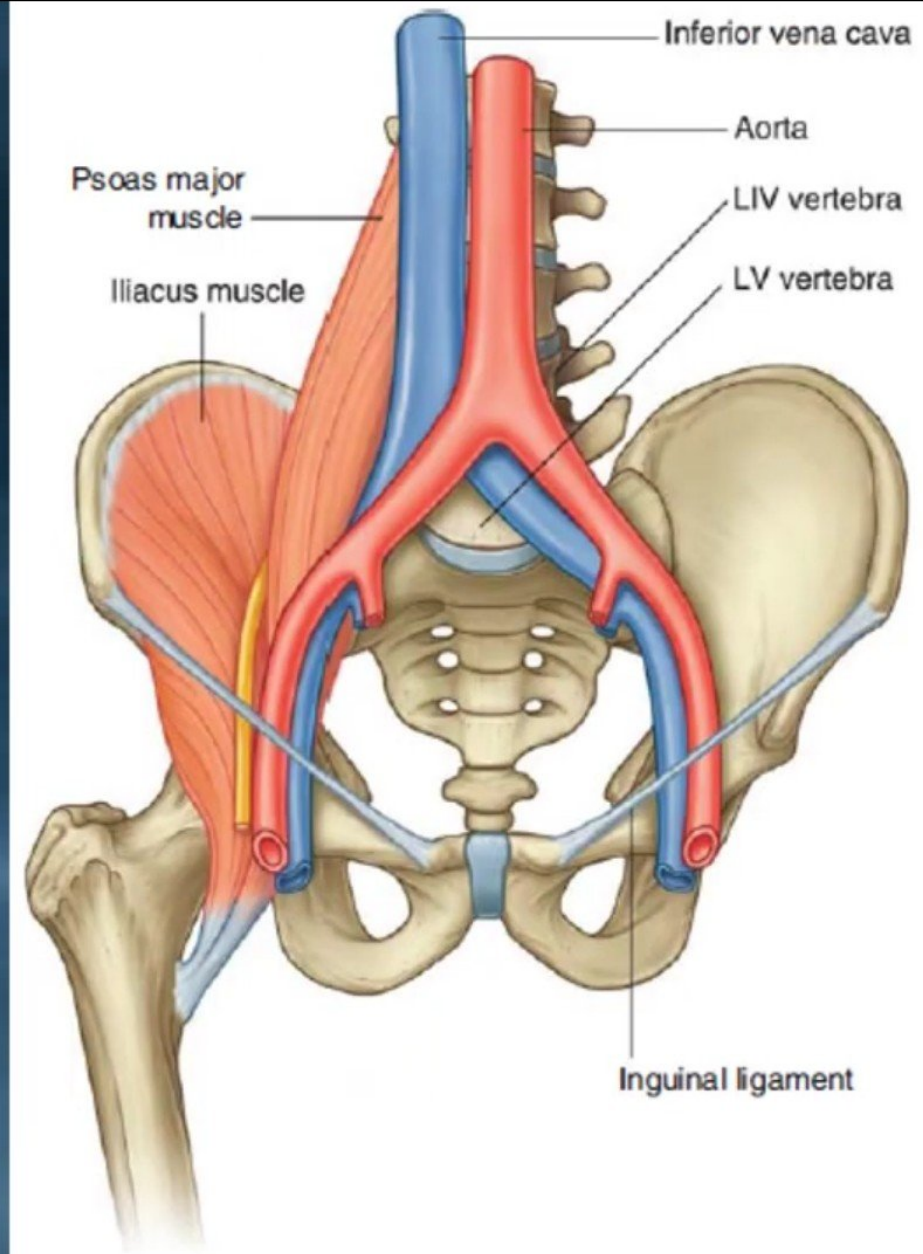
- The pelvic inlet open directly into the abdomen and the structures pass between the abdomen and pelvis through it.
- The peritoneum lining the abdominal cavity is continuous with the peritoneum in the pelvis.
- The infection can spread freely between both regions.
- The bladder extends superiorly from the pelvic cavity into abdominal cavity.
- During pregnancy the uterus extend freely from pelvic cavity into the abdominal cavity.



**The
abdominal
cavity is
continuous
with the
pelvic cavity**

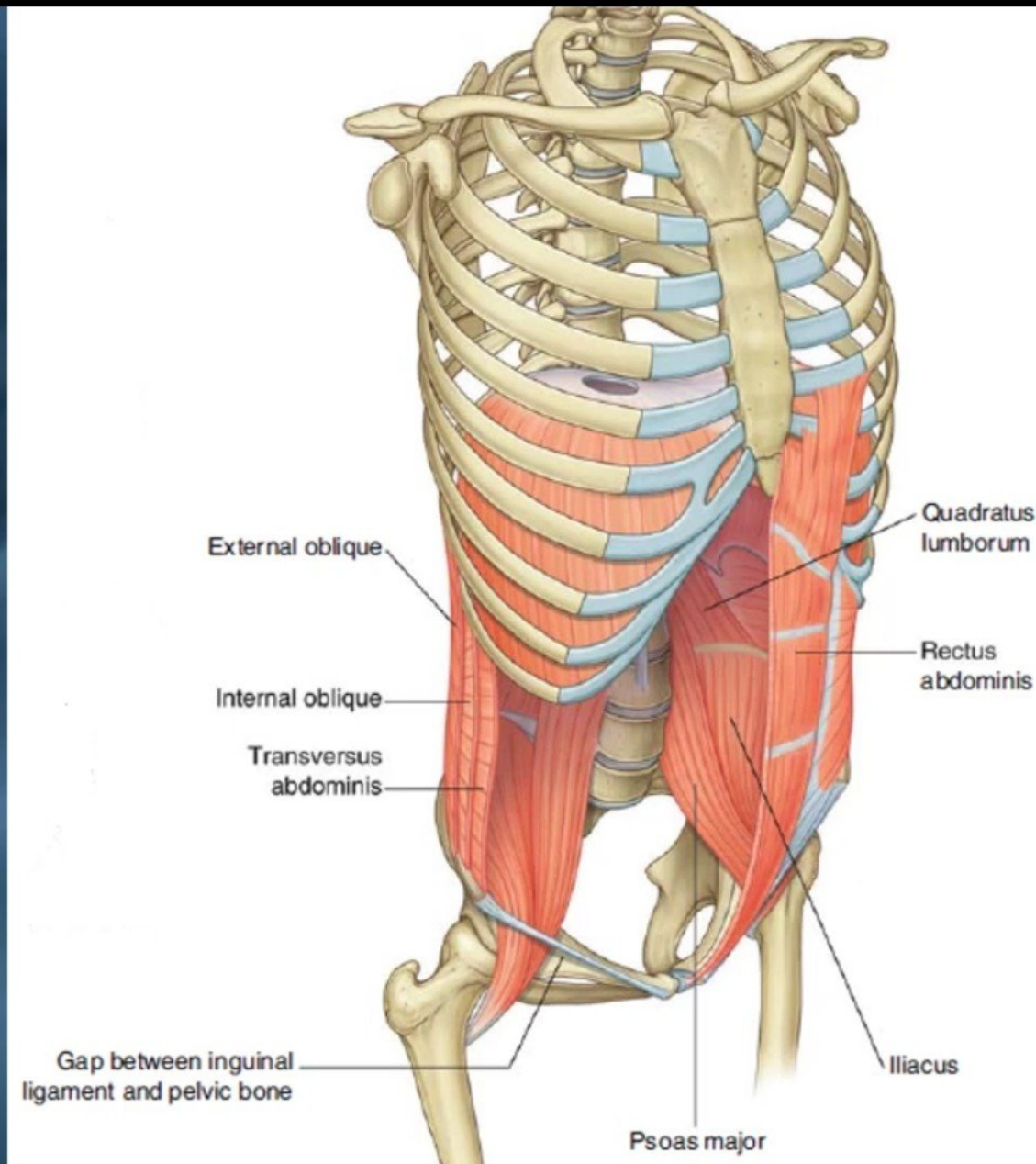


**The abdominal cavity is
continuous with the
pelvic cavity**

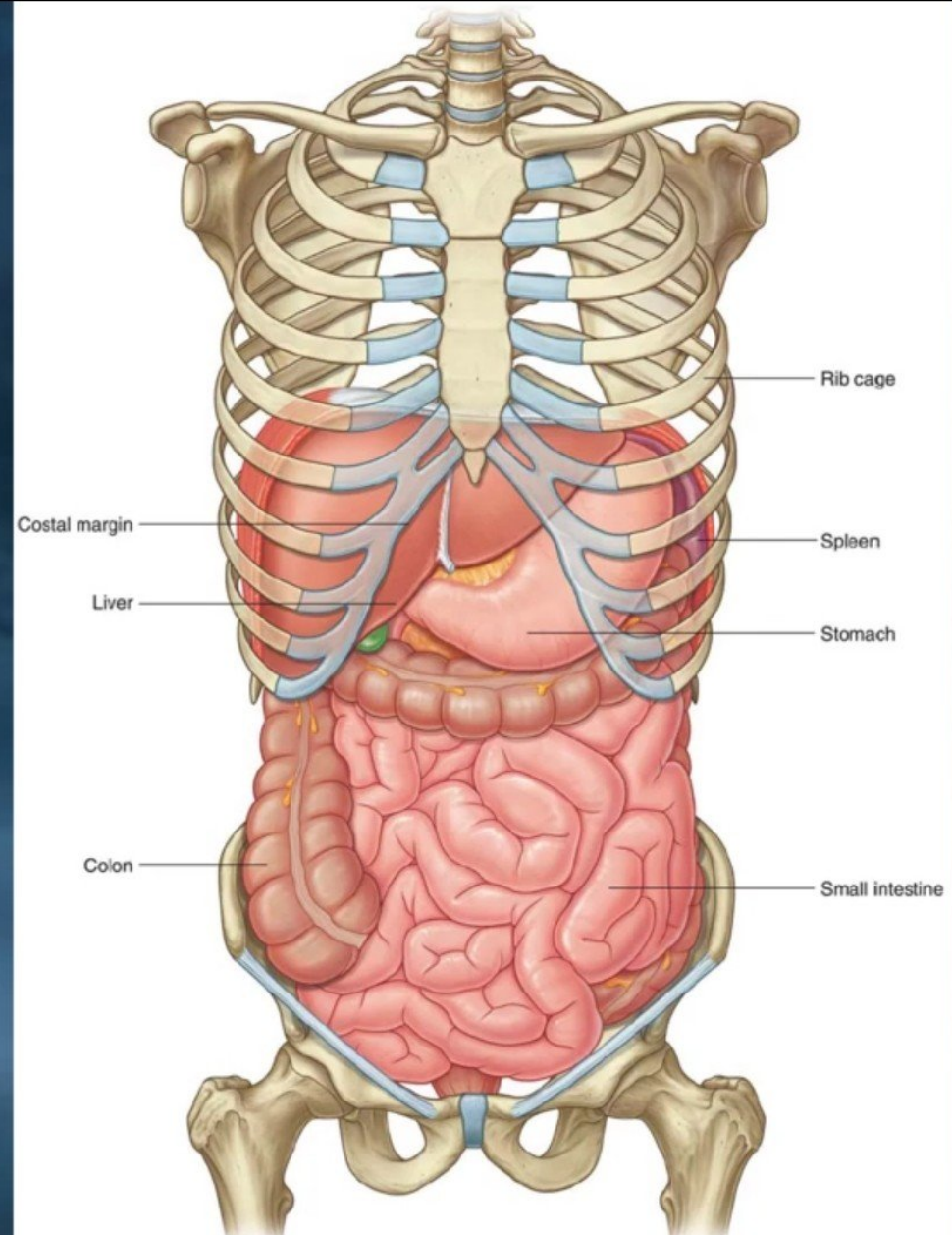


The component parts of abdominal walls

- The muscles make up the rest of abdominal wall.
- Lateral to the vertebral column, **the quadratus lumborum, psoas major** and **iliacus muscles** reinforce the posterior abdominal wall.
- Lateral abdominal wall are formed by three layer of muscles: **transversus abdominis, internal oblique** and **external oblique**.
- Anteriorly found the segmented muscles (the **rectus abdominis**)

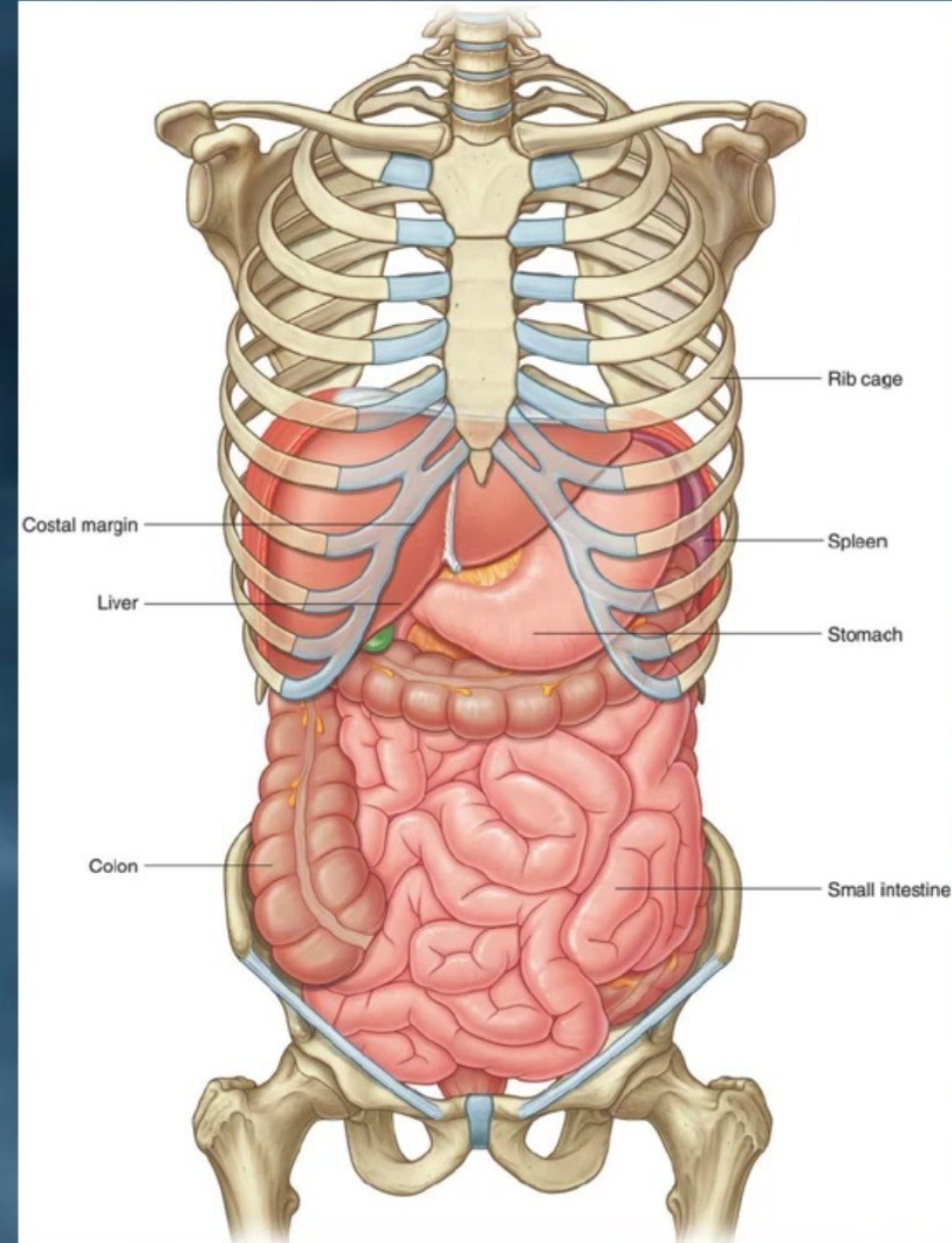


Abdominal Cavity

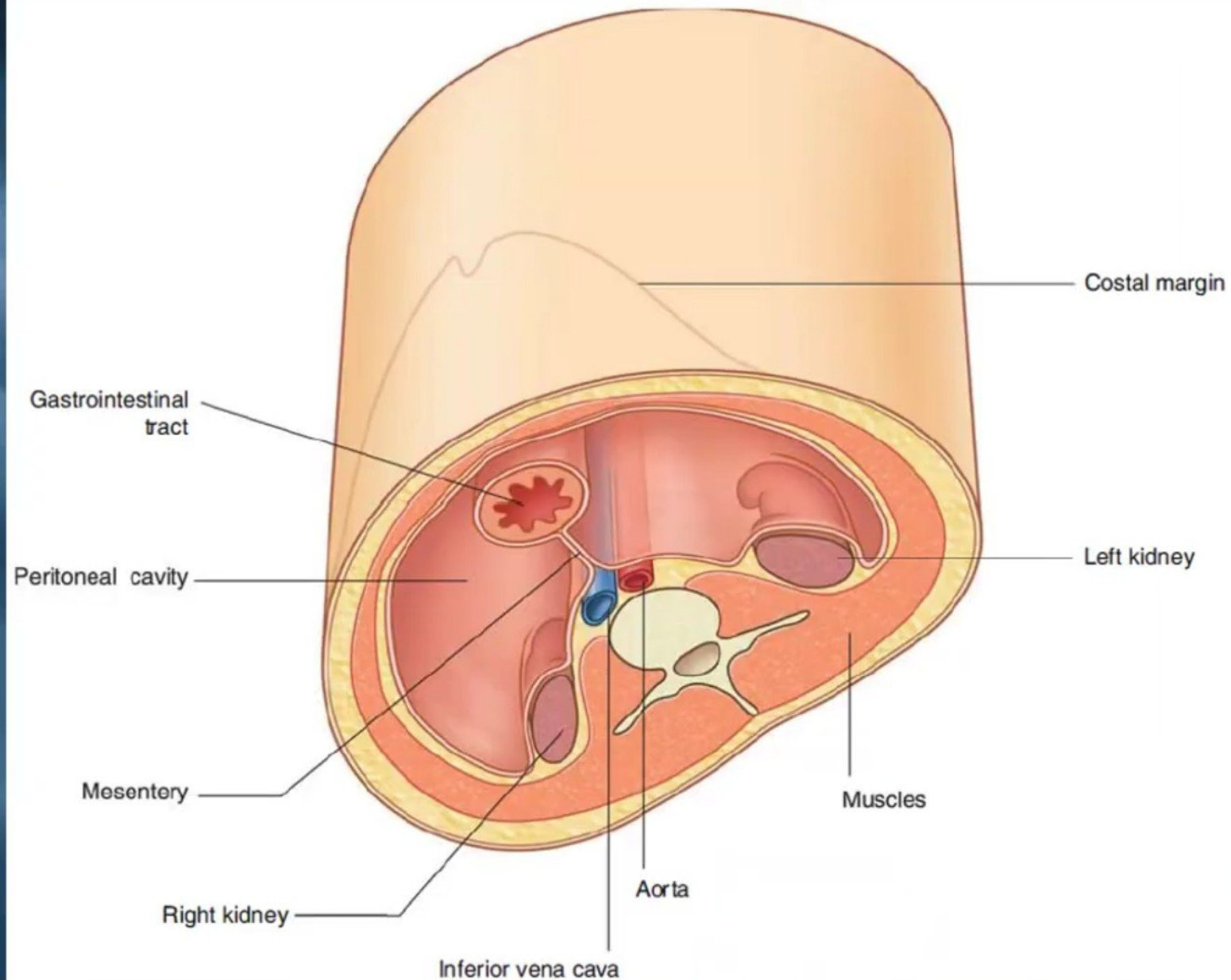


The abdominal Viscera

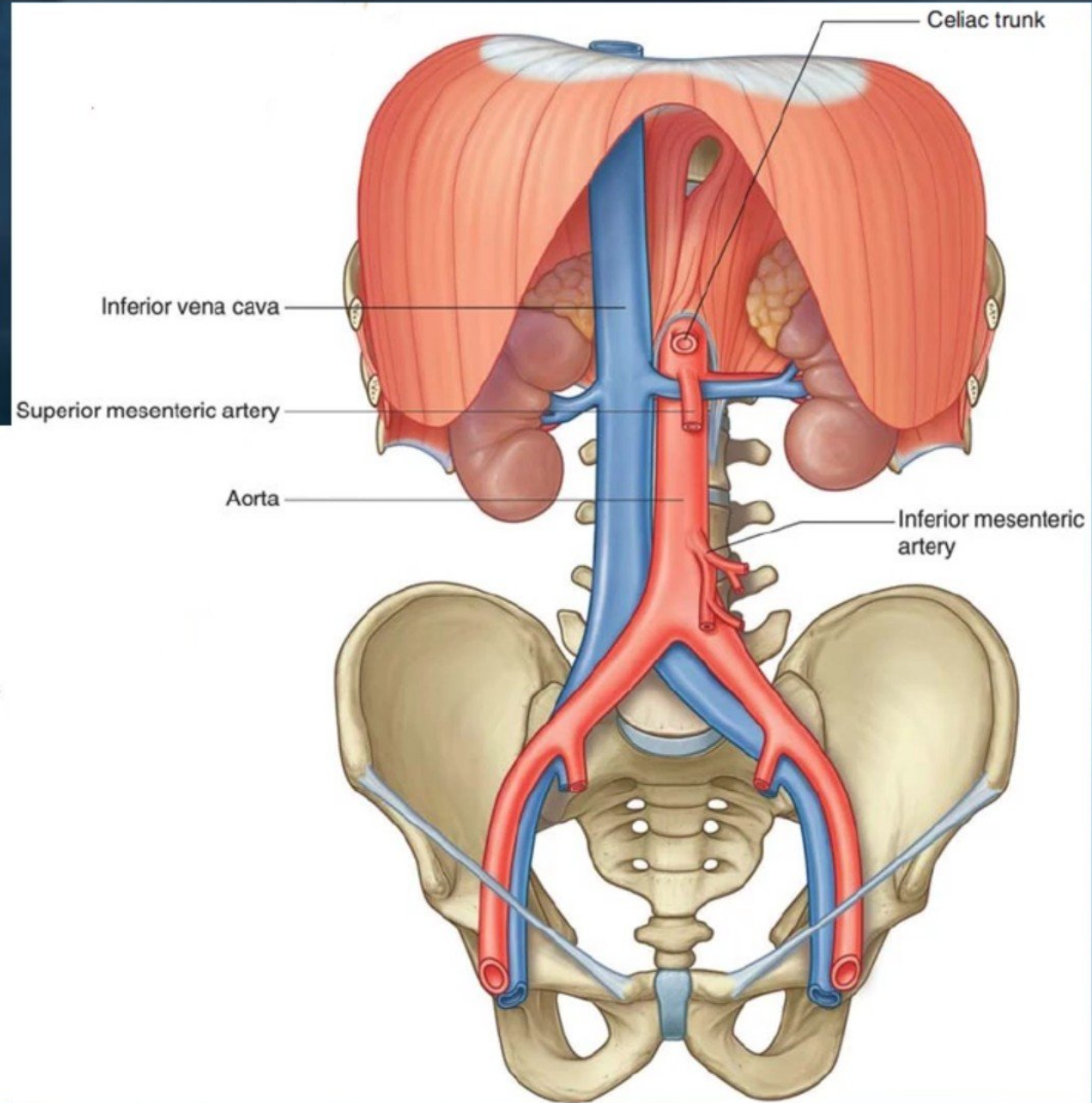
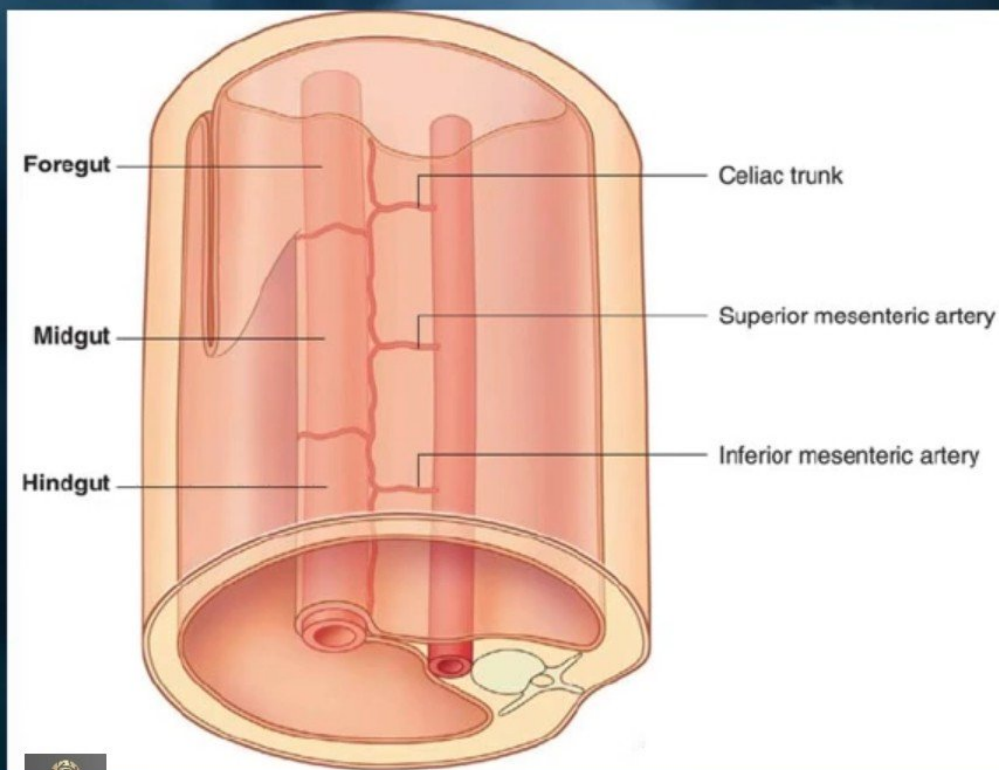
- The major elements of the gastrointestinal system including: the esophagus, stomach, small and large intestines, liver, pancreas and gallbladder.
- The spleen.
- The components of the urinary system including : Kidneys and ureters
- The suprarenal gland.
- And major neurovascular structures.



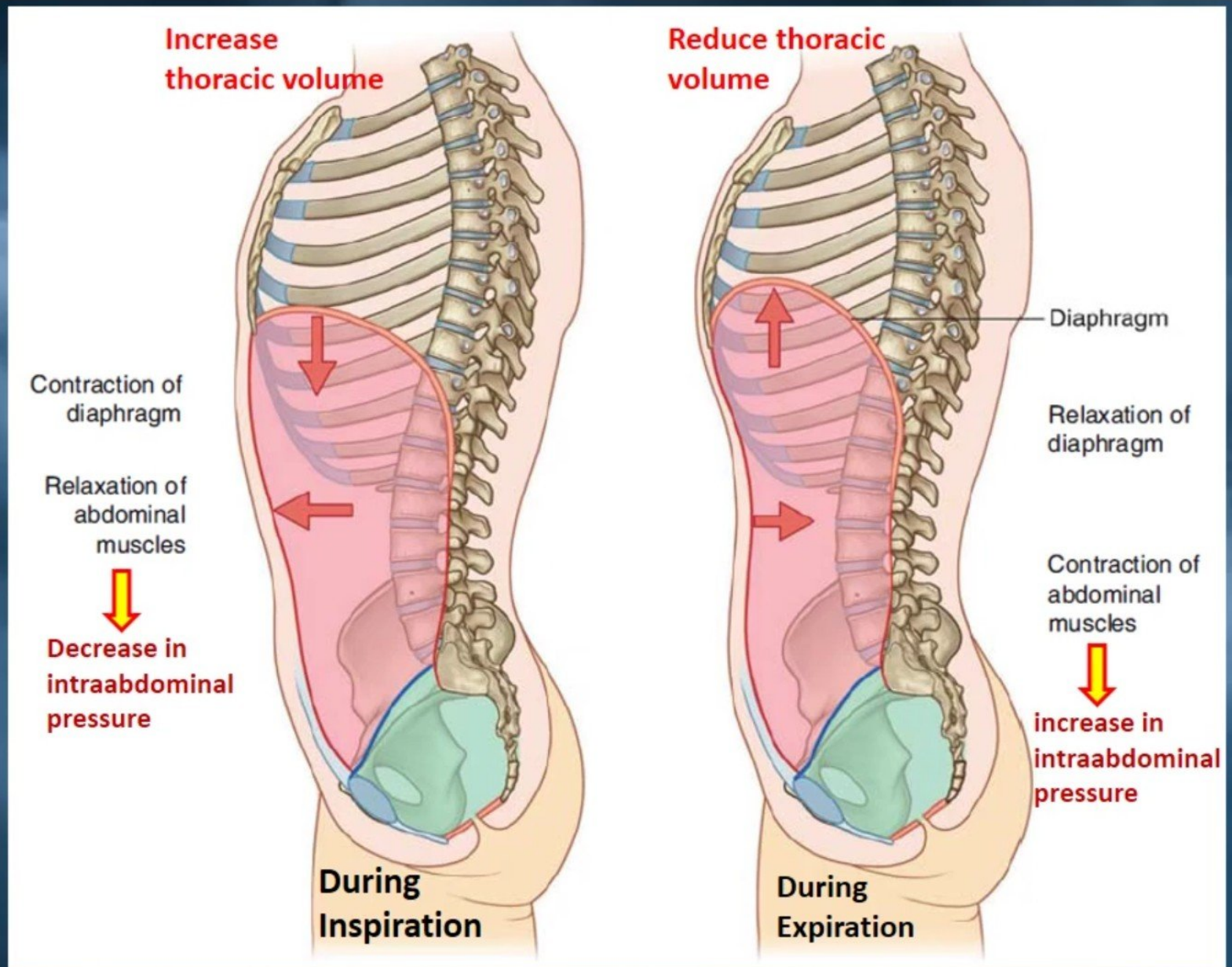
The peritoneal cavity and peritoneum



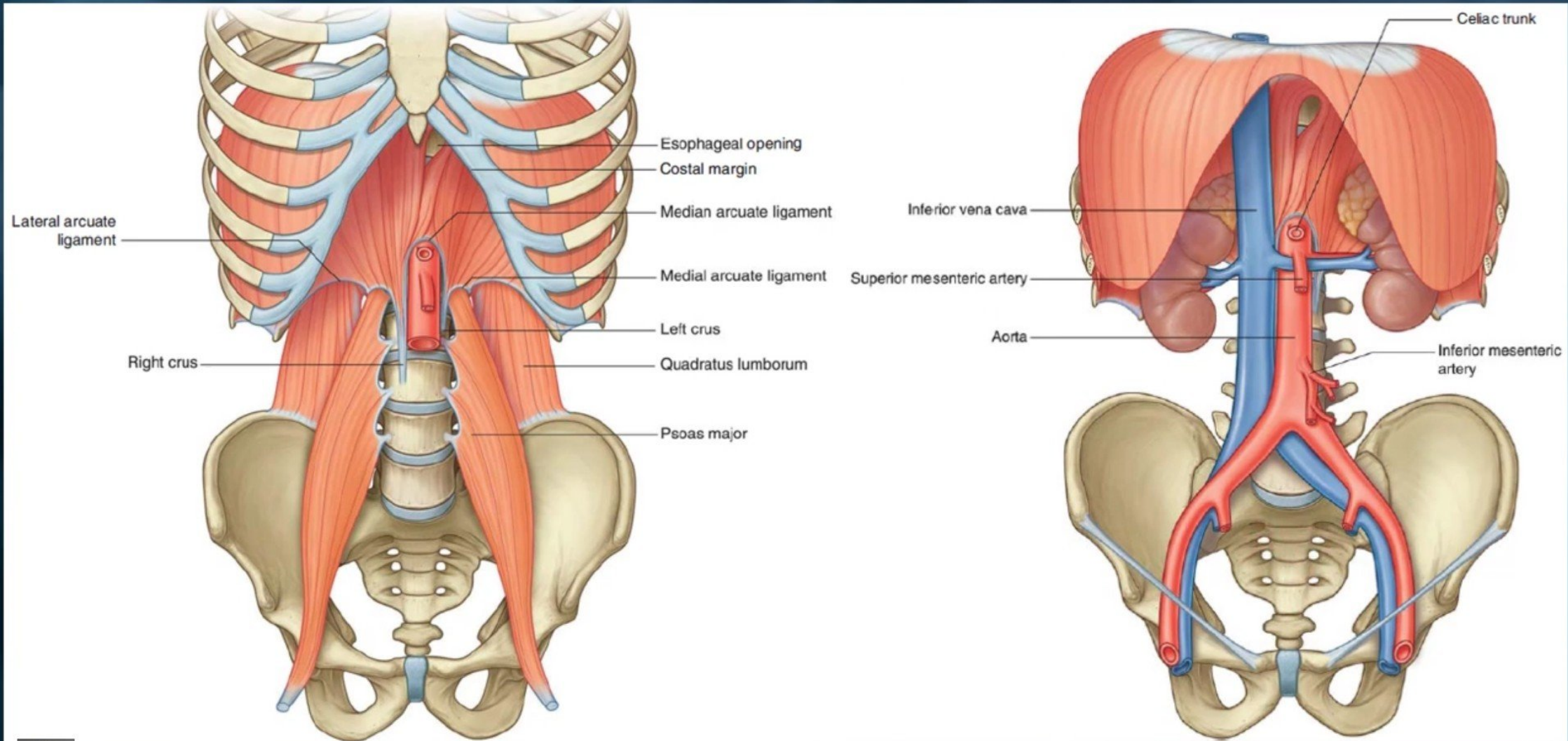
Arterial supply of abdominal organs



Abdomen assist in breathing



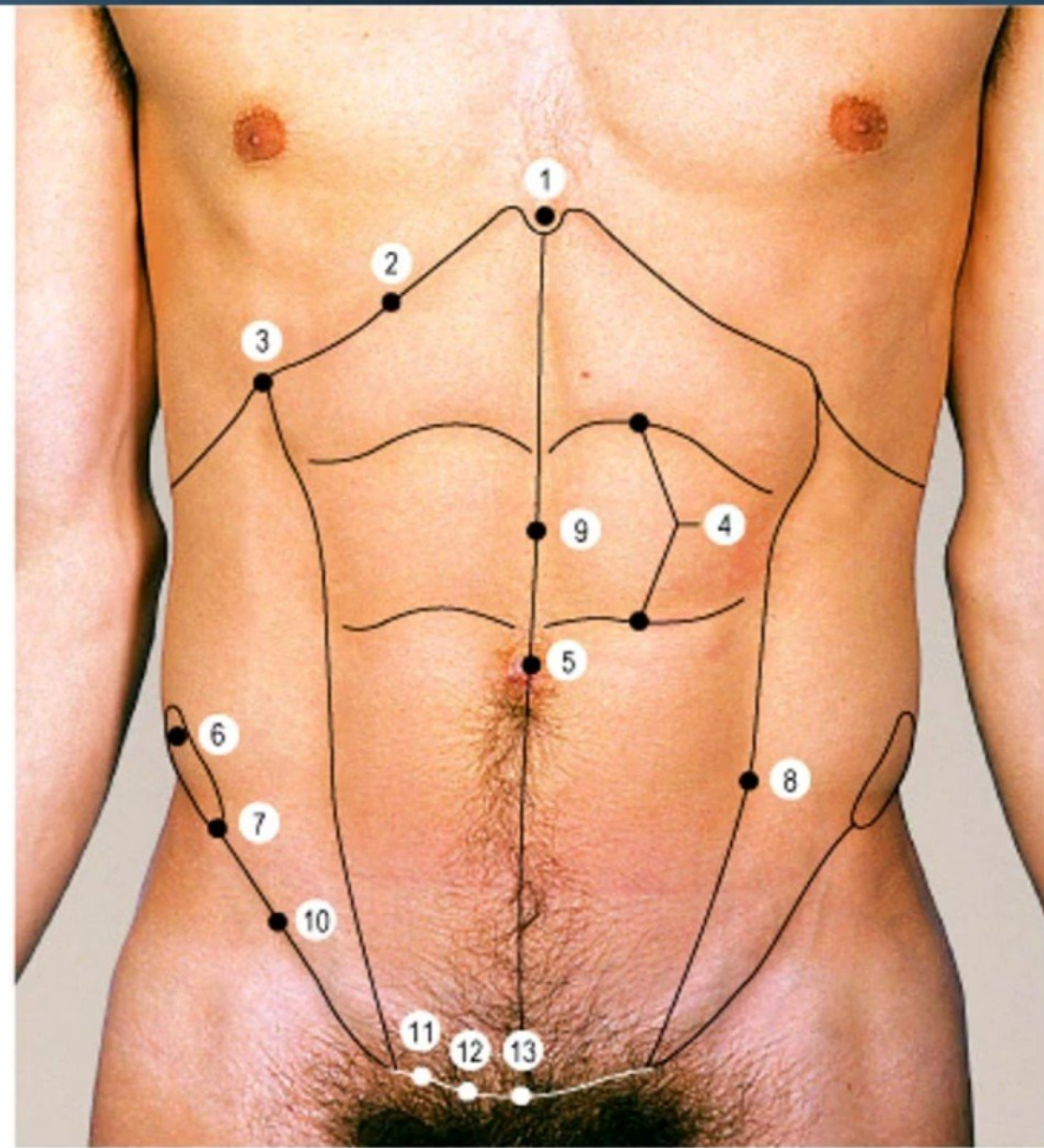
Posterior Abdominal Wall



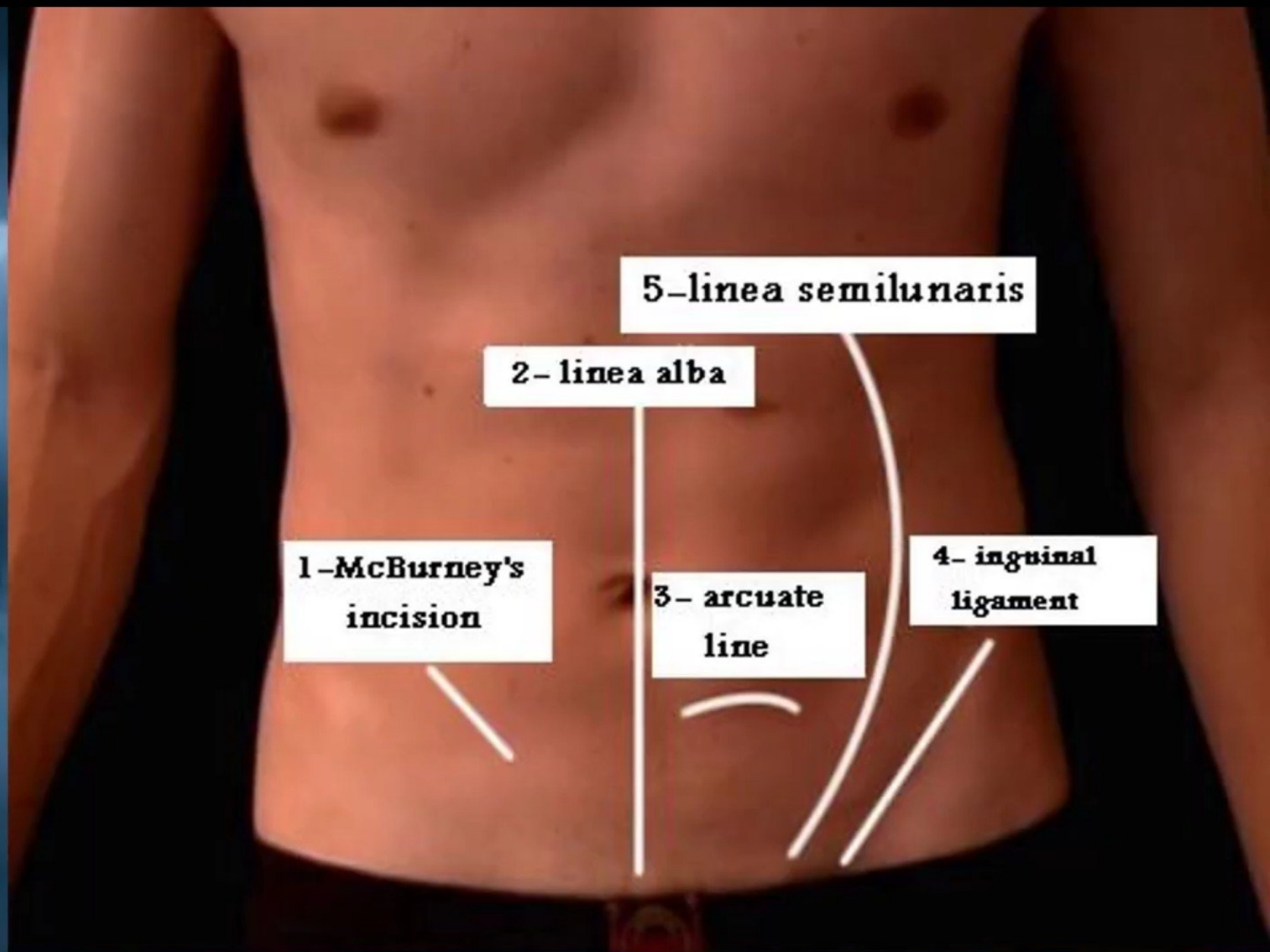
Boundaries of the Abdomen

Boundaries

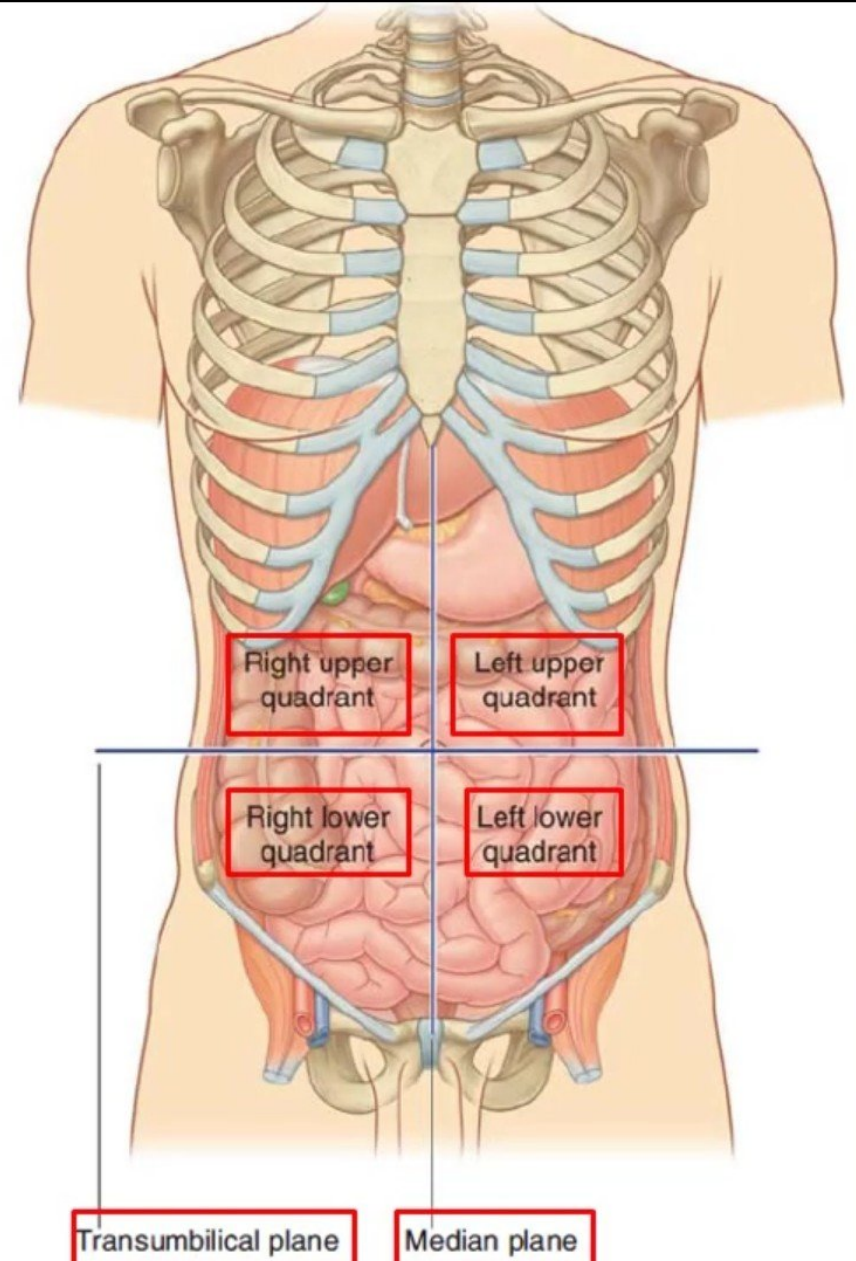
- **Superiorly:** Xiphoid process, lower border of costal arch, 11th and 12th ribs, vertebra T12.
- **Inferiorly:** superior border of pubic symphysis, pubic crest, pubic tubercle, fold of inguinal canal, anterior superior iliac spine, iliac crest, spinous process of L5.



Anatomical landmark



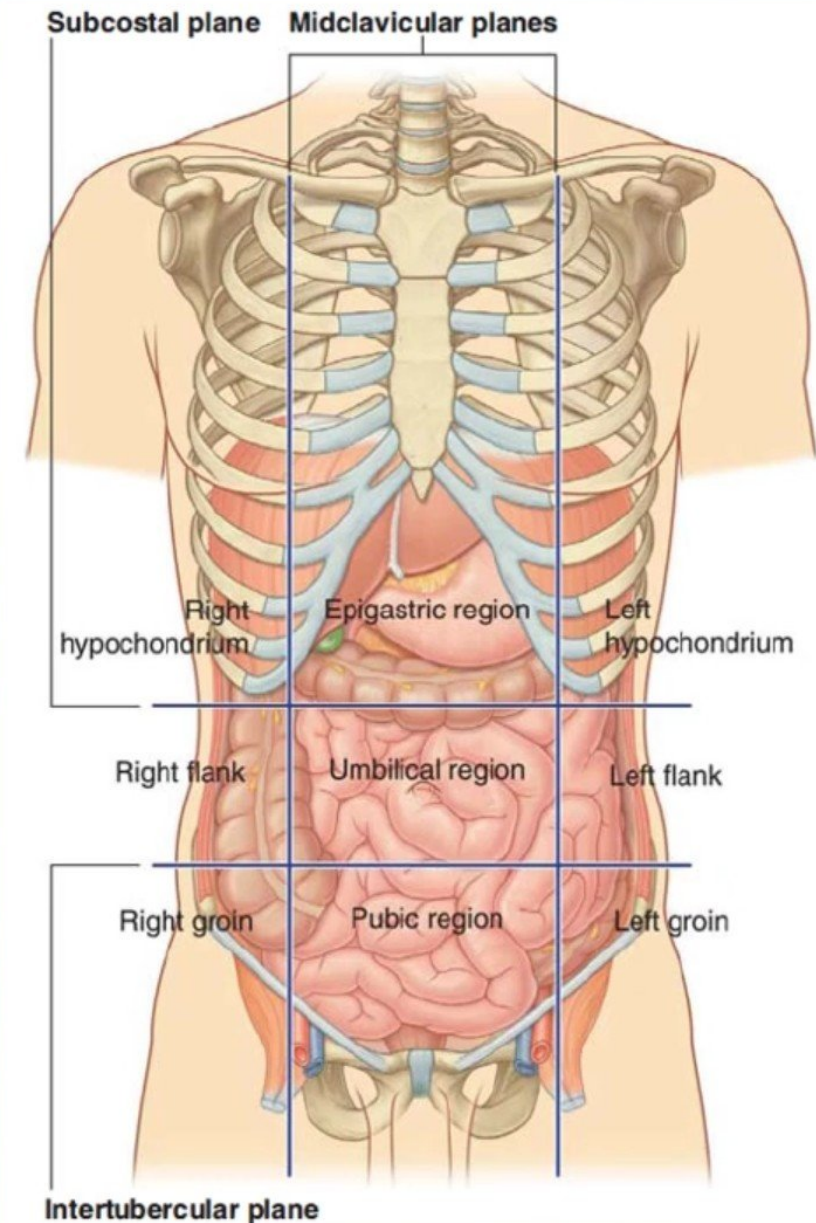
It is divided by
vertical and
horizontal
planes through
the umbilicus
into four
quadrants



The nine region of the abdomen

The nine region :

- **The transpyloric plane (sub costal plane),** also known as **Addison's plane**, is an imaginary **horizontal plane**, located halfway between the suprasternal notch of the manubrium and the upper border of the symphysis pubis at the level of the first lumbar vertebrae, L1.
- **The intertubercular** is lower transverse line midway between passing through the iliac tubercles and behind, its **plane** cuts the body of the fifth lumbar vertebra.



The Anterior Abdominal Wall Functions

- Form strong expandable support.
- Protect the abdominal viscera from injury such as low blow in boxing.
- Compress the abdominal content.
- Helps to maintain or increase the intraabdominal pressure.
- Assist in breathing.
- Move the trunk and help to maintain posture.

Thanks